

Loay Ahmed

Cairo, Egypt | +20 1011969733 | loayahmed304@gmail.com | linkedin.com/in/loayahmed304

EDUCATION

Cairo University

Faculty of Engineering, Computer Department – GPA 3.8

Cairo, Egypt

Expected Aug 2027

EXPERIENCE

stdlib-js

Open-Source Contributor

Cairo, Egypt

Jan 2026 – Present

- Working mainly on migrating Numpy and Python array APIs to stdlib, implementing basic functions kernels and user-facing wrappers.
- Developing packages in C and/or JavaScript, optimizing for performance for critical kernels.

Siemens DISW

DVT Software Engineering Intern

Cairo, Egypt

Oct 2025 – Dec 2025

- Managed C/C++ and TCL projects.
- Applied modern C++ techniques, design patterns, OOP, algorithms, and data structures.

Youverse Learning

FastAPI Backend Intern

Remote

Aug 2025 – Oct 2025

- Managed a FastAPI backend service for an internal employee expenses tracker application.
- Contributed to a FastAPI backend service for the main website, integrating vector databases, semantic searching, DRM, video streaming services, and payments.

TCCD - Cairo University

.NET Backend Intern

Remote

Aug 2025 – Oct 2025

- Built a .NET backend for an HR system handling QR codes and attendance.
- Contributed to the backend of the main website for the annual Cairo University job fair.

PROJECTS

Sockets in C | C, Sockets API, Process Management, Signals

- TCP & UDP-based socket servers and clients, handling connections in separate processes.

Machine Fault Recognition System | Python, Machine Learning, Audio Processing, Docker

Spring 2026

- Developed an end-to-end DL pipeline to classify machine operating states from audio, engineering robust preprocessing modules to handle background noise and silence removal.
- Optimized models for high-accuracy and low-latency inference under strict parameter constraints, and containerized the testing environment using Docker to ensure reliable reproducibility.

“Raven” Infrastructure (Demo) | Docker, Nginx, GitHub Actions, Prometheus, Grafana

- Engineered a multi-environment infrastructure (Production, Staging, Testing) using Docker Compose and Nginx, featuring a highly-available production setup with auto-scaling, a 3-replica backend, and WebSocket/SSL support.
- Designed comprehensive CI/CD pipelines via GitHub Actions to automate semantic versioning, Docker builds, code quality checks, and E2E nightly testing, integrating Watchtower for automated staging deployments and Dozzle/Prometheus/Grafana for real-time system monitoring.

Pipelined RISC Processor | System Design, Computer Architecture, VHDL, Python

- Designed a Von-Neumann 5-stage pipelined processor in VHDL, with a custom Python assembler.

Realtime Background Replacement | Python, Image Processing, OpenCV

- Real-time video stream background replacement, tracking the user as foreground and substituting custom background images.

Go Rate Limiter | Golang, Docker, Rate Limiting

- Fully customizable standalone rate limiting service, defining per-route limits with various algorithms.

Yappuccino | WebSockets, WebRTC, Nginx, Docker, Node.js, React, TS, MongoDB

- Real-time text, voice, and video chat app. Containerized, hosted on EC2 droplet, and automated via GitHub Actions.

Mark Together | *Socket.IO, Node.js, MongoDB, TS, React, Tailwind, Docker*

- Live-collaboration, real-time Markdown editor hosted on private EC2 Droplet managed via Nginx.

Search Engine | *Java, Spring Boot, NoSQL, MongoDB, SEO*

- Custom search engine with real-time crawled data yielding sub-0.5 second search times.

“Sayeh fe Misr” | *Node.js, PostgreSQL, Design Patterns, React.js*

- Online database containing information and ratings about Egyptian tourist places.

Alien Invasion | *C++, Data Structures*

- War simulation incorporating specialized fighting techniques.

Custom Shell | *C, Pipeline, PC, Shell, Linux*

- Custom shell program offering interactive and non-interactive execution with basic shell capabilities.

Digit Recognition | *Python, TensorFlow, ML, FastAPI, TS*

- Simple digit recognition web app using a self-trained neural network.

RESEARCH & ARTICLES

- Debugging and Memory Profiling in C++ – Article Access
- Nginx’s Master-Worker Model: How It Handles Requests Efficiently – Medium Article
- Semantic Searching: What Is It and Its Flow – Medium Article
- WebRTC: Breaking Down WebRTC – Medium Article
- WebSockets: The technology behind them and how they work – Medium Article

COURSES & CERTIFICATES

- **Machine Learning Specialization:** 3 courses by Andrew Ng, Coursera (2025)
- **Docker & Kubernetes: The Practical Guide** by Maximilian Schwarzmüller (2025)
- **Backend Engineering: Introduction to all aspects** by Hussein Nasser (2025)
- **Multiplayer Games Programming: Architecting Networked Games** by Josh Glazer (2026)
- **SQL and PostgreSQL: The Complete Developer’s Guide** by Stephen Grider (2025)
- **Node.js, Express, MongoDB & More: Complete API Bootcamp** by Jonas Schmedtmann (2025)
- **The Complete JavaScript Course** by Jonas Schmedtmann (2024)
- **Google Data Analytics Certificate** – Coursera (2022)

ACTIVITIES

IEEE CUSB

Human Resources Specialist [HR]

- Responsible for managing computer science basics, problem solving, and web workshops.

Cairo, Giza

Oct 2023 – Feb 2024

TECHNICAL SKILLS

Programming Languages: C/C++, C#, Python, Javascript, Typescript, Java, Golang

Networking & Realtime Tech: WebSockets, WebRTC, Nginx

Frameworks & Libraries: React.js, Node.js, Express.js, Nest.js, Qt (C++), OpenCV, OpenAL Soft

Databases: MongoDB, PostgreSQL

Tools & Platforms: Git, Github, GitHub Actions, Docker, Dockerhub, Kubernetes, Postman, Arduino